A1

$\mathbb{F}$ will represent either $\mathbb{R}$ or $\mathbb{C}$

(a) $V = \{(v_x, v_y, 0)\}$

Let $a = (a_x, a_y, 0)$ and $b = (b_x, b_y, 0)$ and $\lambda \in \mathbb{F}$.

$a + b = (a_x + b_x, a_y + b_y, 0) \in V$

$\lambda a = (\lambda a_x, \lambda a_y, 0) \in V$

$\Rightarrow V$ is a vector space.

$\dim V = 2$ with $\{(1,0), (0,1)\}$ an orthonormal basis.

(b) $W = \{(v_x, v_y, 1)\}$

W is not a vector space because

$v = (v_x, v_y, 1)$ and $w = (w_x, w_y, 1)$ both in W

but $v + w = (v_x + w_x, v_y + w_y, 2)$ is not in W.

(c) $X = \{(v_x, v_x, v_x)\}$

Let $a = (a_x, a_x, a_x)$, $b = (b_x, b_x, b_x)$ and $\lambda \in \mathbb{F}$.

$a + b = (a_x + b_x, a_x + b_x, a_x + b_x) \in X$

$\lambda a = (\lambda a_x, \lambda a_x, \lambda a_x) \in X$

$\Rightarrow X$ is a vector space. A basis for it is $(1,1,1)$, and therefore $\dim(X) = 1$.

A2

$$\mathbb{C}_N[X] = \left\{ a_{N-1}X^{N-1} + \cdots + a_1 X + a_0 \mid a_0, a_1, \ldots, a_{N-1} \in \mathbb{C} \right\}$$

(a) Let $v = \sum_{i=0}^{N-1} v_i X^i$ and $w = \sum_{i=0}^{N-1} w_i X^i$ both in $\mathbb{C}_N[X]$

Then $v + w = \sum_{i=0}^{N-1} (v_i + w_i) X^i \in \mathbb{C}_N[X]$ and, for $\lambda \in \mathbb{C}$

$\lambda v = \sum_{i=0}^{N-1} (\lambda v_i) X^i \in \mathbb{C}_N[X]$

$\Rightarrow \mathbb{C}_N[X]$ is a vector space. A basis for it is

$\{1, X, X^2, \ldots, X^{N-1}\}$ and therefore $\dim(\mathbb{C}_N[X]) = N$.

(b) $\mathbb{C}_N^{\text{even}}[X] = \left\{ \sum_{\substack{i=0 \\ i \text{ even}}}^{N-1} a_i X^i \mid a_i \in \mathbb{C} \right\}$

A sum of even polynomials is an even polynomial of degree at most the greatest of their degrees. And a non-zero multiple of an even polynomial, is an even polynomial of the same degree.

$\Rightarrow \mathbb{C}_N^{\text{even}}[X]$ is a vector space.

A basis for it is $\{1, X^2, X^4, \ldots, X^{N-2}\}$ if N is even,

and $\{1, X^2, X^4, \ldots, X^{N-1}\}$ if N is odd

$$\Rightarrow \dim\left(\mathbb{C}_N^{\text{even}}[X]\right) = \begin{cases} \frac{N}{2} & \text{if } N \text{ even} \\ \frac{N+1}{2} & \text{if } N \text{ odd} \end{cases}$$

(c) It is not a vector space, because summing 2 such polynomes gives a polynome with leading coefficient equal to 2, and therefore not an element of the set.

(d) $V = \left\{ \sum_{i=0}^{N-1} v_i x^i \;\middle|\; v_0 + v_1 + \cdots + v_1 = 0 \right\}$

Let $a = \sum_{i=0}^{N-1} a_i x^i$ and $b = \sum_{i=0}^{N-1} b_i x^i$ in V.

$(a+b)(x) = \sum_{i=0}^{N-1} (a_i + b_i) x^i$

and $(a+b)(1) = \sum_{i=0}^{N-1} (a_i + b_i) = \sum_{i=0}^{N-1} a_i + \sum_{i=0}^{N-1} b_i = 0 + 0 = 0$

For $\lambda \in \mathbb{C}$,

$\lambda a(x) = \sum_{i=0}^{N-1} (\lambda a_i) x^i$

$\Rightarrow \lambda a(1) = \sum_{i=0}^{N-1} \lambda a_i = \lambda \sum_{i=0}^{N-1} a_i = 0$

$\Rightarrow V$ is a vector space. A basis for it is $\{1, x, x^2, \cdots, x^{N-2}\}$

Once we fix the coefficients of all but the x^{N-1} term, we deduce its coefficient using the constraint $\sum_{i=0}^{N-1} a_i = 0$.

Therefore $\dim(V) = N-1$.

(e) Not a vector space because the sum of 2 such polynomials would have a value of 2 at $x = 0$.

A.3

Let $B = \{e_1, e_2, \dots, e_n\}$ be a basis for V.

Let $(a_1, \dots, a_n)$ and $(b_1, \dots, b_n)$ be two sets of components for the same vector v with respect to B.

$$\Rightarrow \begin{cases} (1)\ v = a_1 e_1 + a_2 e_2 + \dots + a_n e_n \\ (2)\ v = b_1 e_1 + b_2 e_2 + \dots + b_n e_n \end{cases}$$

$(1) - (2)$: $0 = (a_1 - b_1) e_1 + \dots + (a_n - b_n) e_n$

But because B is a basis, any linear combination of its elements can only be 0 if all coefficients are 0.

$\Rightarrow a_1 = b_1$ and $a_2 = b_2$ and $\dots$ and $a_n = b_n$.

So there is only 1 unique representation of v with respect to B.

Q.E.D

A4

$|e_1\rangle = (1+i)\hat{i} + \hat{j} + i\hat{k} = ((1+i), 1, i)$

$|e_2\rangle = i\hat{i} + 3\hat{j} + \hat{k} = (i, 3, 1)$

$|e_3\rangle = 28\hat{j} = (0, 28, 0)$

$\langle e_1|e_1\rangle = |1+i|^2 + 1 + \|i\|^2 = 2 + 1 + 1 = 4$

$$|e_1'\rangle = \left(\frac{1+i}{2}, \frac{1}{2}, \frac{i}{2}\right)$$

$|f_2\rangle := |e_2\rangle - \langle e_1'|e_2\rangle |e_1'\rangle$

$\langle e_1'|e_2\rangle = \left(\frac{1-i}{2}\cdot i\right) + \frac{1}{2}\cdot 3 - \frac{i}{2}\cdot 1 = 2$

$\Rightarrow |f_2\rangle = (i, 3, 1) - (1+i, 1, i) = (-1, 2, 1-i)$

$$|e_2'\rangle = \frac{|f_2\rangle}{\sqrt{\langle f_2|f_2\rangle}} = \frac{(-1)\hat{i} + 2\hat{j} + (1-i)\hat{k}}{\sqrt{1+4+2}} = \left(-\frac{1}{\sqrt{7}}, \frac{2}{\sqrt{7}}, \frac{1-i}{\sqrt{7}}\right)$$

$|f_3\rangle = |e_3\rangle - \langle e_1'|e_3\rangle |e_1'\rangle - \langle e_2'|e_3\rangle |e_2'\rangle$

$\langle e_1'|e_3\rangle = \frac{28}{2} = 14 \qquad \langle e_2'|e_3\rangle = \frac{56}{\sqrt{7}} = 8\sqrt{7}$

$|f_3\rangle = (0, 28, 0) - 14\cdot\left(\frac{1+i}{2}, \frac{1}{2}, \frac{i}{2}\right) - 8\sqrt{7}\left(-\frac{1}{\sqrt{7}}, \frac{2}{\sqrt{7}}, \frac{1-i}{\sqrt{7}}\right)$

$= (1-7i, 5, -8+i)$

$$|e_3'\rangle = \frac{|f_3\rangle}{\sqrt{\langle f_3|f_3\rangle}}$$

$\langle f_3|f_3\rangle = 50 + 25 + 65 = 140$

$$\Rightarrow |e_3'\rangle = \left(\frac{1-7i}{2\sqrt{35}}, \frac{5}{2\sqrt{35}}, \frac{-8+i}{2\sqrt{35}}\right)$$

A.5

We want to prove: $|\langle\alpha|\beta\rangle|^2 \leq \langle\alpha|\alpha\rangle\langle\beta|\beta\rangle$.

- Case $|\alpha\rangle = |0\rangle$

$$\langle 0|\beta\rangle = \langle\beta|0\rangle^* = \langle\beta|(|0\rangle + |0\rangle)^* = \langle\beta|0\rangle^* + \langle\beta|0\rangle^*$$

$$= \langle 0|\beta\rangle + \langle 0|\beta\rangle$$

$$\Rightarrow \langle 0|\beta\rangle = 0$$

And the inequality holds.

- Case $|\alpha\rangle \neq |0\rangle$

We define

$$|\gamma\rangle = |\beta\rangle - \frac{\langle\alpha|\beta\rangle}{\langle\alpha|\alpha\rangle}|\alpha\rangle$$

$$0 \leq \langle\gamma|\gamma\rangle = \langle\gamma|\left(|\beta\rangle - \frac{\langle\alpha|\beta\rangle}{\langle\alpha|\alpha\rangle}|\alpha\rangle\right)$$

$$= \langle\gamma|\beta\rangle - \frac{\langle\alpha|\beta\rangle}{\langle\alpha|\alpha\rangle}\langle\gamma|\alpha\rangle$$

$$= \left[\langle\beta|\left(|\beta\rangle - \frac{\langle\alpha|\beta\rangle}{\langle\alpha|\alpha\rangle}|\alpha\rangle\right)\right]^* - \frac{\langle\alpha|\beta\rangle}{\langle\alpha|\alpha\rangle}\left[\langle\alpha|\left(|\beta\rangle - \frac{\langle\alpha|\beta\rangle}{\langle\alpha|\alpha\rangle}|\alpha\rangle\right)\right]^*$$

$$= \langle\beta|\beta\rangle - \frac{|\langle\alpha|\beta\rangle|^2}{\langle\alpha|\alpha\rangle} - \frac{\langle\alpha|\beta\rangle}{\langle\alpha|\alpha\rangle}\langle\beta|\alpha\rangle + \frac{\langle\alpha|\beta\rangle}{\langle\alpha|\alpha\rangle}\langle\beta|\alpha\rangle$$

$$= \langle\beta|\beta\rangle - \frac{|\langle\alpha|\beta\rangle|^2}{\langle\alpha|\alpha\rangle}$$

$$\Rightarrow |\langle\alpha|\beta\rangle|^2 \leq \langle\alpha|\alpha\rangle\langle\beta|\beta\rangle$$

QED

A.6

$|\alpha\rangle = (1+i,\ 1,\ i)$ and $|\beta\rangle = (4-i,\ 0,\ 2-2i)$

$\langle\alpha|\alpha\rangle = 2 + 1 + 1 = 4$

$\langle\beta|\beta\rangle = 17 + 0 + 8 = 25$

$\langle\alpha|\beta\rangle = (3-5i) + (-2-2i) = 1 - 7i$

$\Rightarrow |\langle\alpha|\beta\rangle|^2 = 1 + 49 = 50$

$\Rightarrow \cos\theta = \sqrt{\frac{50}{4 \cdot 25}} = \frac{1}{\sqrt{2}} \Rightarrow \boxed{\theta = \frac{\pi}{4}}$

A.7

$\| (|\alpha\rangle + |\beta\rangle) \|^2$

$= \|\alpha\|^2 + \|\beta\|^2 + \langle\alpha|\beta\rangle + \langle\beta|\alpha\rangle$

$= \|\alpha\|^2 + \|\beta\|^2 + 2\,\mathrm{Re}(\langle\alpha|\beta\rangle)$

$\leq \|\alpha\|^2 + \|\beta\|^2 + 2\,|\langle\alpha|\beta\rangle|$

$\leq \|\alpha\|^2 + \|\beta\|^2 + 2\,\|\alpha\| \cdot \|\beta\|$ (by C.S. inequality)

$= (\|\alpha\| + \|\beta\|)^2$

$\Rightarrow \| (|\alpha\rangle + |\beta\rangle) \| \leq \|\alpha\| + \|\beta\|$ $\boxed{\text{Q.E.D}}$

A.8

(a) $A+B = \begin{pmatrix} 1 & 1 & 0 \\ 2 & 1 & 3 \\ 3i & 3-2i & 4 \end{pmatrix}$

(b) $AB = \begin{pmatrix} -3 & 1+3i & 3i \\ 4+3i & 9 & 6-2i \\ 6i & 6-2i & 6 \end{pmatrix}$

(c) $BA = \begin{pmatrix} 0 & 0 & 0 \\ 2 & 0 & 3 \\ 6+3i & -3i & 12 \end{pmatrix}$

$[A,B] = AB - BA = \begin{pmatrix} -3 & 1+3i & 3i \\ 2+3i & 9 & 3-2i \\ -6+3i & 6+i & -6 \end{pmatrix}$

(d) $\tilde{A} = \begin{pmatrix} -1 & 2 & 2i \\ 1 & 0 & -2i \\ i & 3 & 2 \end{pmatrix}$

(e) $A^* = \begin{pmatrix} -1 & 1 & -i \\ 2 & 0 & 3 \\ -2i & 2i & 2 \end{pmatrix}$

(f) $A^\dagger = \begin{pmatrix} -1 & 2 & -2i \\ 1 & 0 & 2i \\ -i & 3 & 2 \end{pmatrix}$

(g) $\det(B) = \begin{vmatrix} 2 & 0 & -i \\ 0 & 1 & 0 \\ i & 3 & 2 \end{vmatrix} = \begin{vmatrix} 2 & -i \\ i & 2 \end{vmatrix} = 3$

(h) $B^{-1} = \frac{1}{3}\begin{pmatrix} 2 & -3i & i \\ 0 & 3 & 0 \\ -i & -6 & 2 \end{pmatrix}$

$$B B^{-1} = \frac{1}{3}\begin{pmatrix} 2 & 0 & -i \\ 0 & 1 & 0 \\ i & 3 & 2 \end{pmatrix}\begin{pmatrix} 2 & -3i & i \\ 0 & 3 & 0 \\ -i & -6 & 2 \end{pmatrix} = \frac{1}{3}\begin{pmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{pmatrix} = I$$

$$\det(A) = \begin{vmatrix} -1 & 1 & i \\ 2 & 0 & 3 \\ 2i & -2i & 2 \end{vmatrix} = 6i + 4 - 4 - 6i = 0$$

and so A^{-1} does not exist.

A.9

(a) $$Aa = \begin{pmatrix} -1 & 1 & i \\ 2 & 0 & 3 \\ 2i & -2i & 2 \end{pmatrix}\begin{pmatrix} i \\ 2i \\ 2 \end{pmatrix} = \begin{pmatrix} 3i \\ 6+2i \\ 6 \end{pmatrix}$$

(b) $$a^\dagger b = \begin{pmatrix} -i & -2i & 2 \end{pmatrix}\begin{pmatrix} 2 \\ 1-i \\ 0 \end{pmatrix} = -2i - 2i(1-i) = -2-4i$$

(c) $$\tilde{a} B b = \begin{pmatrix} i & 2i & 2 \end{pmatrix}\begin{pmatrix} 2 & 0 & -i \\ 0 & 1 & 0 \\ i & 3 & 2 \end{pmatrix}\begin{pmatrix} 2 \\ 1-i \\ 0 \end{pmatrix} = \begin{pmatrix} 4i & 6+2i & 5 \end{pmatrix}\begin{pmatrix} 2 \\ 1-i \\ 0 \end{pmatrix}$$

$$= 8i + (6+2i)(1-i) = 8i + 6 - 6i + 2i + 2 = 8 + 4i$$

(d) $$a b^\dagger = \begin{pmatrix} i \\ 2i \\ 2 \end{pmatrix}\begin{pmatrix} 2 & 1+i & 0 \end{pmatrix} = \begin{pmatrix} 2i & -1+i & 0 \\ 4i & -2+2i & 0 \\ 4 & 2+2i & 0 \end{pmatrix}$$

A.10

(a) $B = \underbrace{\frac{B+\tilde{B}}{2}}_{S} + \underbrace{\frac{B-\tilde{B}}{2}}_{A}$

(b) $B = \underbrace{\frac{B+B^*}{2}}_{R} + \underbrace{\frac{B-B^*}{2}}_{M}$

(c) $B = \underbrace{\frac{B+B^\dagger}{2}}_{H} + \underbrace{\frac{B-B^\dagger}{2}}_{K}$

A.11

- **Proof of $\widetilde{ST} = \tilde{T}\tilde{S}$**

Let $S = S^{p\times q}$ and $T = T^{q\times m}$

The i,j-th element of $\widetilde{ST}$ is the j,i-th element of ST

$$\left(\widetilde{ST}\right)_{i,j} = (ST)_{j,i} = \sum_{k=1}^{q} S_{j,k}\, T_{k,i}$$

The i,j-th element of $\tilde{T}\cdot\tilde{S}$ is

$$\left(\tilde{T}\cdot\tilde{S}\right)_{i,j} = \sum_{k=1}^{q} \tilde{T}_{i,k}\,\tilde{S}_{k,j} = \sum_{k=1}^{q} T_{k,i}\, S_{j,k}$$

QED

- **Proof of $(ST)^\dagger = T^\dagger S^\dagger$**

$$(ST)^\dagger = \left(\widetilde{ST}\right)^* = \left(\tilde{T}\cdot\tilde{S}\right)^* = T^\dagger S^\dagger$$

QED

- **Proof of $(ST)^{-1} = T^{-1}S^{-1}$**

$$\left(T^{-1}S^{-1}\right)(ST) = T^{-1}\left(S^{-1}S\right)T = T^{-1}T = I$$

and

$$(ST)\left(T^{-1}S^{-1}\right) = S\left(TT^{-1}\right)S^{-1} = SS^{-1} = I$$

QED

- Proof that product of unitary is unitary

Let A, B be unitary.

$$(AB)^{-1} = B^{-1} A^{-1} = B^{\dagger} A^{\dagger} = (AB)^{\dagger}$$ QED

- Let A, B be hermitian. We want $(AB)^{\dagger} = AB$.
We know
$$(AB)^{\dagger} = B^{\dagger} A^{\dagger} = BA.$$
Therefore AB is hermitian if $[A, B] = 0$.

- Let $A = B = I$. They are both unitary.
But
$$(A+B)^{-1} = (2I)^{-1} = \frac{1}{2} I$$
while
$$(A+B)^{\dagger} = 2I$$
So the sum of unitary matrices need not be unitary.

- Let A, B be hermitian.
$$(A+B)^{\dagger} = A^{\dagger} + B^{\dagger} = A+B$$
So the sum of hermitian matrices is hermitian.

A.12

Let A be an $n \times n$, unitary matrix.

$\Rightarrow A^{\dagger} = A^{-1}$

Let $R = \{r_1, r_2, \dots, r_n\}$ be the set of rows of A and

$C = \{c_1, c_2, \dots, c_n\}$ the set of its columns.

$$A = \begin{pmatrix} - r_1 - \\ - r_2 - \\ \vdots \\ - r_n - \end{pmatrix} = \begin{pmatrix} | & | & & | \\ c_1 & c_2 & \cdots & c_n \\ | & | & & | \end{pmatrix}$$

$$I = A A^{\dagger} = \begin{pmatrix} - r_1 - \\ - r_2 - \\ \vdots \\ - r_n - \end{pmatrix} \begin{pmatrix} | & | & & | \\ r_1^* & r_2^* & \cdots & r_n^* \\ | & | & & | \end{pmatrix}$$

$$\Rightarrow \left(- r_i -\right) \begin{pmatrix} | \\ r_j^* \\ | \end{pmatrix} = \delta_{ij}$$

So the rows are an orthonormal set. We also have

$$I = A^{\dagger} A = \begin{pmatrix} - c_1^* - \\ - c_2^* - \\ \vdots \\ - c_n^* - \end{pmatrix} \begin{pmatrix} | & | & & | \\ c_1 & c_2 & \cdots & c_n \\ | & | & & | \end{pmatrix}$$

$$\Rightarrow \left(- c_i^* -\right) \begin{pmatrix} | \\ c_j \\ | \end{pmatrix} = \delta_{i,j}$$ QED

A.13

- Let $A = A^\dagger$

$$\det(A) = \det(A^\dagger) = \det\left((\tilde{A})^*\right) = \left(\det(\tilde{A})\right)^* = \left(\det(A)\right)^*$$

$\Rightarrow \det(A)$ is real. QED

- Let $A^\dagger = A^{-1}$

$$1 = \det(I) = \det(A A^\dagger) = \det(A) \cdot \det(A^\dagger)$$

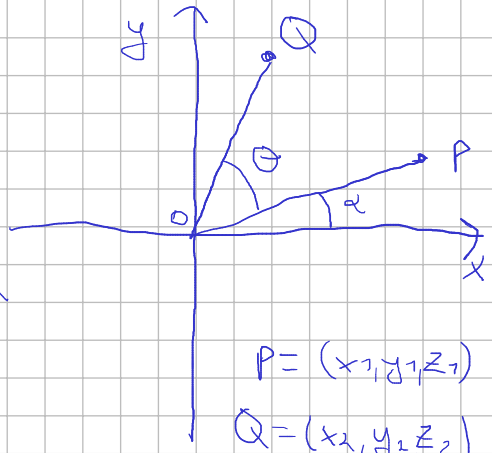
$$= \det(A) \cdot \left(\det(A)\right)^* = |\det(A)|^2$$ QED

- Let $\tilde{A} = A^{-1}$

From the existence A^{-1}, we know $\det(A) \neq 0$.

$$1 = \det(I) = \det(A\tilde{A}) = \left(\det(A)\right)^2 \Rightarrow \det(A) = \pm 1$$ QED

A 14



(a) Let R be the distance from the origin to P, the point where P ends up after the rotation, α the angle that OP makes with the positive x-axis.

$$\begin{cases} x_1 = R\cos\alpha \\ y_1 = R\sin\alpha \end{cases}$$

$$\begin{cases} x_2 = R\cos(\alpha+\theta) = R\cos\alpha\cos\theta - R\sin\alpha\sin\theta = x_1\cos\theta - y_1\sin\theta \\ y_2 = R\sin(\alpha+\theta) = R\sin\alpha\cos\theta + R\cos\alpha\sin\theta = y_1\cos\theta + x_1\sin\theta \end{cases}$$

The rotation matrix M is therefore

$$\begin{pmatrix} x_2 \\ y_2 \\ z_2 \end{pmatrix} = M \begin{pmatrix} x_1 \\ y_1 \\ z_1 \end{pmatrix} = \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ y_1 \\ z_1 \end{pmatrix} \qquad (*)$$

(b) We want the rotation matrix around $(1,1,1)$. One way of getting it from the previous result, is:

- Choose an orthonormal basis such that its 3rd element is $(1,1,1)$.
- Determine the transformation matrix S from the original basis to this new one.
- Applying S to any vector, followed by R_0 (the rotation matrix from part (a)), followed by S^{-1}, amounts to the rotation we want, i.e.: $R = S^{-1} R_0 S$

For this problem, the original basis B is $\{(1,0,0), (0,1,0), (0,0,1)\}$. One possible choice for the new basis B' is

$$B' = \left\{ \left(\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}, 0\right), \left(\frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}, -\frac{2}{\sqrt{6}}\right), \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right) \right\}$$

Because B' is orthonormal, the elements of S are the dot products of the elements of B' with those of B.

$$S = \begin{pmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{6}} & -\frac{2}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \end{pmatrix}$$

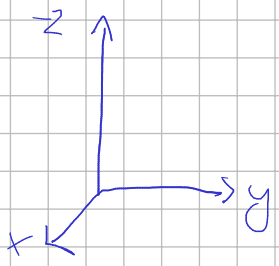
I showed in my notes of the theory of this appendix that such transformation matrices are unitary $\Rightarrow \quad S^{-1} = S^{\dagger}$.

$\Rightarrow R = S^{-1} R_0 S$ I plugged 120° for θ in (*)

$$= \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ 0 & -\frac{2}{\sqrt{6}} & \frac{1}{\sqrt{3}} \end{pmatrix} \begin{pmatrix} -\frac{1}{2} & -\sqrt{3}/2 & 0 \\ \sqrt{3}/2 & -\frac{1}{2} & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{6}} & -\frac{2}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \end{pmatrix}$$

$$= \begin{pmatrix} 0 & -\frac{2}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{6}} & -\frac{2}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \end{pmatrix} = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$$

(c) The vector $v = \begin{pmatrix} v_x \\ v_y \\ v_z \end{pmatrix}$ becomes $\begin{pmatrix} v_x \\ v_y \\ -v_z \end{pmatrix}$ after reflection through the xy-plane

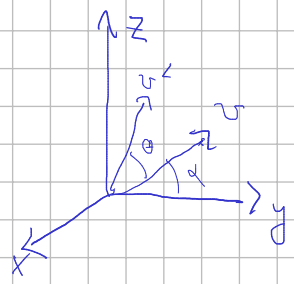


$$\Rightarrow S = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$$

(d) I proved in the theory notes of this appendix that the transformation matrix between orthonormal bases is unitary. As all the matrices above are such transformation matrices, and they are all real, it follows they are all orthogonal.
The determinant of the matrix from (a) is $\cos^2\theta + \sin^2\theta = 1$. For (b) it is 1. And for (c) it is -1.

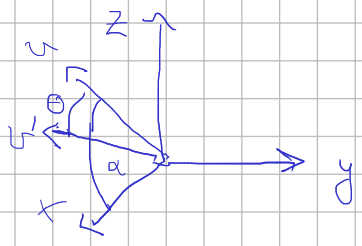
A.15

$$\begin{cases} v_y = v\cos\alpha \\ v_z = v\sin\alpha \end{cases}$$



$$\begin{cases} v'_y = v\cos(\alpha+\theta) = v\cos\alpha\cos\theta - v\sin\alpha\sin\theta = v_y\cos\theta - v_z\sin\theta \\ v'_z = v\sin(\alpha+\theta) = v\sin\alpha\cos\theta + v\cos\alpha\sin\theta = v_z\cos\theta + v_y\sin\theta \end{cases}$$

$$\Rightarrow T_x = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{pmatrix}$$



Similarly,

$$\begin{cases} v_x = v\cos\alpha \\ v_z = v\sin\alpha \end{cases}$$

$$v'_x = v\cos(\alpha-\theta) = v\cos\alpha\cos\theta + v\sin\alpha\sin\theta = v_x\cos\theta + v_z\sin\theta$$

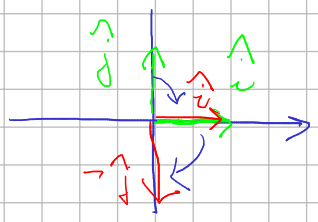
$$v'_z = v\sin(\alpha-\theta) = v\sin\alpha\cos\theta - v\cos\alpha\sin\theta = v_z\cos\theta - v_x\sin\theta$$

$$\Rightarrow T_y = \begin{pmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{pmatrix}$$

$$(\hat{i}, \hat{j}, \hat{k}) \rightarrow (\hat{j}, -\hat{i}, \hat{k}) \Rightarrow S = \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

- The basis vector $\hat{i} = (1,0,0)$ is mapped to $-\hat{j} = (0,-1,0)$
- The basis vector $\hat{j} = (0,1,0)$ is mapped to $\hat{i} = (1,0,0)$

As the z component is unchanged, I only draw the xy-plane



This mapping is thus a rotation over the z-axis by 90° clockwise.

$S\,T_x\,S^{-1}$ should represent a rotation over the z-axis over 90° counterclockwise, followed by a rotation over the x-axis over θ, followed by a rotation over the z axis over 90° clockwise.

To see what we expect $S T_x S^{-1}$ to do, note that S^{-1} amounts to the relabeling $x \to -y$, $y \to x$, $z \to z$

Therefore, applying T_x to this boils down to rotating over $-y$ by an angle θ. And the following S undoes the relabeling. So we expect $S T_x S^{-1}$ to be a rotation over the $-y$ axis over an angle θ.

$$S T_x S^{-1} = \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 0 & \cos\theta & -\sin\theta \\ -1 & 0 & 0 \\ 0 & \sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} \cos\theta & 0 & -\sin\theta \\ 0 & 1 & 0 \\ \sin\theta & 0 & \cos\theta \end{pmatrix}$$

Which is exactly what we found for T_y, with θ replaced by $-\theta$.

Similarly, we expect $S T_y S^{-1}$ to be a rotation of θ around the x-axis.

$$S T_y S^{-1} = \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{pmatrix} \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 0 & 1 & 0 \\ -\cos\theta & 0 & -\sin\theta \\ -\sin\theta & 0 & \cos\theta \end{pmatrix} \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{pmatrix} = T_x \quad \text{as expected.}$$

A.16

Thm: Let $A^f = S A^e S^{-1}$, $B^f = S B^e S^{-1}$, and $C^f = S C^e S^{-1}$. If $A^e B^e = C^e$, then $A^f B^f = C^f$

Proof:

$$A^f B^f = S A^e S^{-1} S B^e S^{-1} = S A^e B^e S^{-1} = S C^e S^{-1} = C^f$$

QED

Thm: H^e Hermitian $\Rightarrow$ H^f Hermitian

Proof: Let H^e be Hermitian. I already showed in the theory notes of this appendix that S is unitary.

$$(H^f)^\dagger = (S H^e S^{-1})^\dagger = (S^{-1})^\dagger H^e S^\dagger = S H^e S^{-1} = H^f$$

QED

Thm: Let $B = \{|e_1\rangle, \dots, |e_n\rangle\}$ be an orthonormal basis.

Then: $B' = \{|f_1\rangle, \dots, |f_n\rangle \mid |f_i\rangle = S|e_i\rangle\}$ is orthonormal basis $\Leftrightarrow$ S is unitary.

Proof:

$\Rightarrow$

$$S_{ij} = \langle f_i | f_j \rangle = \langle f_i | S | e_j \rangle$$

$$\overset{(1)}{=} \langle e_j | S^\dagger | f_i \rangle^* = \langle e_j | S^\dagger S | e_i \rangle$$

Let $S^\dagger S |e_i\rangle = \sum_{k=1}^{n} (S^\dagger S)_{ki} |e_k\rangle$

$$\Rightarrow S_{ij} = \sum_{k=1}^{n} (S^\dagger S)_{ki} \langle e_j | e_k \rangle$$

$$= (S^\dagger S)_{ji}$$

$\Rightarrow S^\dagger S = I$, so S is unitary.

equality (1) holds because of the theorem $\langle \alpha | S | \beta \rangle = \langle \beta | S^\dagger | \alpha \rangle^*$ which I proved in the theory notes of this appendix.

$\Leftarrow$

$$\langle f_i | f_j \rangle = \langle e_i | S^\dagger S | e_j \rangle = \langle e_i | e_j \rangle = \delta_{ij}$$

Therefore the $|f_i\rangle$'s are orthonormal, and thus linearly independent. As the space is n-dimensional they form a basis.

QED

A.17

Thm: Let T_1, T_2 be $n \times n$ matrices. Then $\mathrm{Tr}(T_1 T_2) = \mathrm{Tr}(T_2 T_1)$

Proof:

$$(T_1 T_2)_{ij} = \sum_{k=1}^{n} T_{1\,ik}\, T_{2\,kj}$$

$$\Rightarrow \mathrm{Tr}(T_1 T_2) = \sum_{i=1}^{n} (T_1 T_2)_{ii} = \sum_{i=1}^{n} \sum_{k=1}^{n} T_{1\,ik}\, T_{2\,ki}$$

$$= \sum_{k=1}^{n} \left(\sum_{i=1}^{n} T_{2\,ki}\, T_{1\,ik} \right) = \sum_{k=1}^{n} (T_2 T_1)_{kk}$$

$$= \mathrm{Tr}(T_2 T_1) \qquad \boxed{\text{QED}}$$

$$\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$

But $\mathrm{Tr}(T_1 T_2 T_3) \neq \mathrm{Tr}(T_2 T_1 T_3)$

counterexample: $T_1 = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$ $T_2 = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$ $T_3 = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$

$T_1 T_2 T_3 = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \Rightarrow \mathrm{Tr}(T_1 T_2 T_3) = 0$

But

$T_2 T_1 T_3 = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \Rightarrow \mathrm{Tr}(T_2 T_1 T_3) = 1$

A.18

$$T = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$$ I use $\sin\theta \equiv s$, $\cos\theta \equiv t$

$$0 = \begin{vmatrix} t-\lambda & -s \\ s & t-\lambda \end{vmatrix} = (t-\lambda)^2 + s^2 = \lambda^2 - 2\lambda t + 1$$

$$\Rightarrow \lambda = \frac{2t \pm \sqrt{4t^2-4}}{2} = t \pm \sqrt{t^2-1}$$

There are only real solutions when $t = \cos\theta = \pm 1$, i.e. when $\theta = k\pi$ with $k \in \mathbb{Z}$. In that case, any non-zero vector is an eigenvector. For $\theta \neq k\pi$, the (complex) eigenvalues are $\lambda = \cos\theta \pm i \sin\theta$

$\lambda = \cos\theta + i \sin\theta$

$$\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} v_x \\ v_y \end{pmatrix} = (\cos\theta + i\sin\theta) \begin{pmatrix} v_x \\ v_y \end{pmatrix}$$

$$\Leftrightarrow \begin{cases} v_x \cos\theta - v_y \sin\theta = v_x \cos\theta + i v_x \sin\theta \\ v_x \sin\theta + v_y \cos\theta = v_y \cos\theta + i v_y \sin\theta \end{cases}$$

$$\Leftrightarrow \begin{cases} -v_y = i v_x & (1) \\ v_x = i v_y & (2) \end{cases}$$

Therefore the eigenvector is $v_1 = \begin{pmatrix} 1 \\ -i \end{pmatrix}$

$\lambda = \cos\theta - i\sin\theta$

$$\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} v_x \\ v_y \end{pmatrix} = (\cos\theta - i\sin\theta) \begin{pmatrix} v_x \\ v_y \end{pmatrix} \Leftrightarrow \begin{cases} v_x \cos\theta - v_y \sin\theta = v_x \cos\theta - i v_x \sin\theta \\ v_x \sin\theta + v_y \cos\theta = v_y \cos\theta - i v_y \sin\theta \end{cases}$$

$$\Leftrightarrow \begin{cases} v_y = i v_x \\ v_x = -i v_y \end{cases}$$

and so the other eigenvector is $v_2 = \begin{pmatrix} 1 \\ i \end{pmatrix}$

The transformation matrix is $S^{-1} = \begin{pmatrix} 1 & 1 \\ -i & i \end{pmatrix}$

$\det(S^{-1}) = 2i$

$$\Rightarrow \overline{S} = \frac{1}{2i}\begin{pmatrix} i & i \\ -1 & 1 \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{i}{2} & -\frac{i}{2} \end{pmatrix} \Rightarrow S = \frac{1}{2}\begin{pmatrix} 1 & i \\ 1 & -i \end{pmatrix}$$

$$STS^{-1} = \frac{1}{2}\begin{pmatrix} 1 & i \\ 1 & -i \end{pmatrix}\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}\begin{pmatrix} 1 & 1 \\ -i & i \end{pmatrix}$$

$$= \frac{1}{2}\begin{pmatrix} \cos\theta + i\sin\theta & -\sin\theta + i\cos\theta \\ \cos\theta - i\sin\theta & -\sin\theta - i\cos\theta \end{pmatrix}\begin{pmatrix} 1 & 1 \\ -i & i \end{pmatrix}$$

$$= \begin{pmatrix} \cos\theta + i\sin\theta & 0 \\ 0 & \cos\theta - i\sin\theta \end{pmatrix} = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}$$

A.19

$$M = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

$$0 = \begin{vmatrix} 1-\lambda & 1 \\ 0 & 1-\lambda \end{vmatrix} = (1-\lambda)^2 \Rightarrow \lambda = 1$$

There is only one eigenvalue

$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} v_x \\ v_y \end{pmatrix} = \begin{pmatrix} v_x \\ v_y \end{pmatrix} \Leftrightarrow \begin{cases} v_x + v_y = v_x \\ v_y = v_y \end{cases} \Rightarrow v_y = 0$$

$$\Rightarrow v = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

There are no 2 linearly independent eigenvectors, and therefore the matrix M cannot be diagonalized.

A.20

Equations (A.72) and (A.73) say

$$(*)\quad 0 = \begin{vmatrix} T_{11}-\lambda & T_{12} & \cdots & T_{1n} \\ T_{21} & T_{22}-\lambda & \cdots & T_{2n} \\ \vdots & & & \\ T_{n1} & T_{n2} & \cdots & T_{nn}-\lambda \end{vmatrix} = C_n\lambda^n + C_{n-1}\lambda^{n-1} + \cdots + C_1\lambda + C_0 = 0$$

The highest order term comes from taking all the terms with λ in $(T_{11}-\lambda)(T_{22}-\lambda)\cdots(T_{nn}-\lambda)$, which gives

$$C_n\lambda^n = (-\lambda)^n = (-1)^n\lambda^n \Rightarrow C_n = (-1)^n$$

To calculate C_{n-1}, I first expand the determinant by its 1st column:

$$0 = (T_{11}-\lambda)\begin{vmatrix} T_{22}-\lambda & T_{23} & \cdots & T_{2n} \\ T_{32} & T_{33}-\lambda & \cdots & T_{3n} \\ \vdots & & & \\ T_{n2} & T_{n3} & \cdots & T_{nn}-\lambda \end{vmatrix} + T_{21}\begin{vmatrix} T_{12} & T_{13} & \cdots & T_{1n} \\ T_{32} & T_{33}-\lambda & \cdots & T_{3n} \\ \vdots & & & \\ T_{n2} & T_{n3} & \cdots & T_{nn}-\lambda \end{vmatrix} +$$

$$+ \cdots + (-1)^{n+1} T_{n1}\begin{vmatrix} T_{12} & T_{13} & \cdots & T_{1n} \\ T_{22}-\lambda & T_{23} & \cdots & T_{2n} \\ T_{(n-1)2} & T_{(n-1)3} & \cdots & T_{(n-1)n} \end{vmatrix}$$

Except for the first term in the sum, all the others have orders of λ of at most λ^{n-2}. Therefore C_n and C_{n-1} are solely determined by the first term.

$$1^{st}\text{ term} = (T_{11}-\lambda)\left[(T_{22}-\lambda)\begin{vmatrix} T_{33}-\lambda & T_{34} & \cdots & T_{3n} \\ T_{43} & T_{44}-\lambda & & T_{4n} \\ \vdots & & & \\ T_{n3} & T_{n4} & \cdots & T_{nn}-\lambda \end{vmatrix} + \begin{array}{c}\text{all terms with} \\ \text{order of } \lambda \text{ at most} \\ n-3\end{array}\right]$$

So, once again, only the 1st term contributes to C_n and C_{n-1}. We see that the contributions for C_n and C_{n-1} only come from

$$(T_{11}-\lambda)(T_{22}-\lambda)\cdots(T_{nn}-\lambda) =$$

$$= (-1)^n\lambda^n$$

$$+ \left[T_{11}(-\lambda)^{n-1} + T_{22}(-\lambda)^{n-1} + \cdots + T_{nn}(-\lambda)^{n-1}\right]$$

+ terms of order of λ of at most $n-2$.

$$\Rightarrow \begin{cases} c_n = (-1)^n \\ c_{n-1} = (-1)^{n-1} \, T_2(T) \end{cases}$$

In order to find c_0, we note that the terms without λ are exactly those that would be left in (*) if we were to plug $\lambda = 0$ into it.

$\Rightarrow c_0 = \det(T)$.

For a 3×3 matrix, we have:

$$\begin{vmatrix} T_{11}-\lambda & T_{12} & T_{13} \\ T_{21} & T_{22}-\lambda & T_{23} \\ T_{31} & T_{32} & T_{33}-\lambda \end{vmatrix} = (T_{11}-\lambda)(T_{22}-\lambda)(T_{33}-\lambda) + T_{12}T_{23}T_{31} + T_{13}T_{21}T_{32} - T_{13}T_{31}(T_{22}-\lambda) - T_{12}T_{21}(T_{33}-\lambda) - (T_{11}-\lambda)T_{23}T_{32}$$

the terms with λ are

$$c_1 \lambda = \left(-T_{11}T_{22} - T_{11}T_{33} - T_{22}T_{33} + T_{13}T_{31} + T_{12}T_{21} + T_{23}T_{32}\right)\lambda$$

A.21

Thm:
Let T be an $n \times n$ matrix, and let $\lambda_1, \dots, \lambda_n$ be the solutions of its characteristic equation.
Then $\det(T) = \lambda_1 \lambda_2 \cdots \lambda_n$ and $Tr(T) = \lambda_1 + \lambda_2 + \cdots + \lambda_n$

Proof:
The characteristic equation is

$$(1) \quad (-1)^n \lambda^n + (-1)^{n-1} Tr(T) \lambda^{n-1} + \sum_{j=1}^{n-2} C_j \lambda^j + \det(T) = 0$$

Using the n solutions of the equation, this can also be written as

$$(2) \quad (-1)^n (\lambda - \lambda_1)(\lambda - \lambda_2) \cdots (\lambda - \lambda_n) = 0$$

The terms without λ are $\det(T)$ in (1) and $\lambda_1 \lambda_2 \cdots \lambda_n$ in (2).

$\Rightarrow \det(T) = \lambda_1 \lambda_2 \cdots \lambda_n$

The terms with λ^{n-1} are $(-1)^{n-1} Tr(T) \lambda^{n-1}$ in (1)
and $(-1)^n \left[-\lambda_1 \lambda^{n-1} - \lambda_2 \lambda^{n-1} - \cdots - \lambda_n \lambda^{n-1} \right]$ in (2)

$\Rightarrow Tr(T) = \lambda_1 + \lambda_2 + \cdots + \lambda_n$ QED

A.22

(a) $M = \begin{pmatrix} 1 & 1 \\ 1 & i \end{pmatrix} \Leftrightarrow M^\dagger = \begin{pmatrix} 1 & 1 \\ 1 & -i \end{pmatrix}$

So that

$MM^\dagger = \begin{pmatrix} 2 & 1-i \\ 1+i & 2 \end{pmatrix}$ and $M^\dagger M = \begin{pmatrix} 2 & 1+i \\ 1-i & 2 \end{pmatrix}$

therefore $[M^\dagger, M] \neq 0$ and so M is not normal.

(b) $\begin{vmatrix} 1-\lambda & 1 \\ 1 & i-\lambda \end{vmatrix} = (1-\lambda)(i-\lambda) - 1 = \lambda^2 - (i+1)\lambda + (i-1) = 0$

$$\Rightarrow \lambda = \frac{i+1 \pm \sqrt{(i+1)^2 - 4(i-1)}}{2} = \frac{i+1 \pm \sqrt{-2i+4}}{2}$$

There are two (unequal) eigenvalues, and therefore its eigenvectors span the space. Therefore M is diagonalizable.

A.23

Thm: $[T_1^e, T_2^e] = 0 \Rightarrow [T_1^b, T_2^b] = 0$

Proof:

Equation (A.64) tells us that $T^b = S T^e S^{-1}$

$$\Rightarrow [T_1^b, T_2^b] = T_1^b T_2^b - T_2^b T_1^b = S T_1^e S^{-1} S T_2^e S^{-1} - S T_2^e S^{-1} S T_1^e S^{-1}$$

$$= S T_1^e T_2^e S^{-1} - S T_2^e T_1^e S^{-1}$$

$$= S [T_1^e, T_2^e] S^{-1}$$

$$= 0$$

QED

A.24

We want to choose the d coefficients in (A.88) such that

$$V\tilde{a}^{(i)} = \nu_i \tilde{a}^{(i)}$$

Using (A.88) and the equations above it, this becomes

$$V\left(d_{1i}\, a^{(1)} + d_{2i}\, a^{(2)}\right) = \nu_i \left(d_{1i}\, a^{(1)} + d_{2i}\, a^{(2)}\right)$$

$$\Leftrightarrow d_{1i}\left(c_{11}\, a^{(1)} + c_{21}\, a^{(2)}\right) + d_{2i}\left(c_{12}\, a^{(1)} + c_{22}\, a^{(2)}\right) = \nu_i \left(d_{1i}\, a^{(1)} + d_{2i}\, a^{(2)}\right)$$

Because $a^{(1)}$ and $a^{(2)}$ are linearly independent, this can only hold if it holds for the coefficients of $a^{(1)}$ and $a^{(2)}$ separately:

$$\begin{cases} c_{11}\, d_{1i} + c_{12}\, d_{2i} = \nu_i\, d_{1i} \\ c_{21}\, d_{1i} + c_{22}\, d_{2i} = \nu_i\, d_{2i} \end{cases}$$

This is exactly (A.90).

A.25

(a) Both A and B are real and symmetric, and therefore hermitian. This means they are both normal, and thus diagonisable.

$$AB = \begin{pmatrix} 1 & 4 & 1 \\ 4 & -2 & 4 \\ 1 & 4 & 1 \end{pmatrix}\begin{pmatrix} 1 & -2 & -1 \\ -2 & 2 & -2 \\ -1 & -2 & 1 \end{pmatrix} = \begin{pmatrix} -8 & 4 & -8 \\ 4 & -20 & 4 \\ -8 & 4 & -8 \end{pmatrix}$$

$$BA = \begin{pmatrix} 1 & -2 & -1 \\ -2 & 2 & -2 \\ -1 & -2 & 1 \end{pmatrix}\begin{pmatrix} 1 & 4 & 1 \\ 4 & -2 & 4 \\ 1 & 4 & 1 \end{pmatrix} = \begin{pmatrix} -8 & 4 & -8 \\ 4 & -20 & 4 \\ -8 & 4 & -8 \end{pmatrix}$$

$$\Rightarrow [A, B] = 0$$

(b)

$$0 = \begin{vmatrix} 1-\lambda & 4 & 1 \\ 4 & -2-\lambda & 4 \\ 1 & 4 & 1-\lambda \end{vmatrix} = -(1-\lambda)^2(2+\lambda) + 16 + 16 + (2+\lambda) - 16(1-\lambda) - 16(1-\lambda)$$

$$= -\lambda^3 + 36\lambda = \lambda(-\lambda^2 + 36)$$

So the eigenvalues are $0, 6, -6$.

The spectrum is non-degenerate.

$\lambda = 0$

$$\begin{pmatrix} 1 & 4 & 1 \\ 4 & -2 & 4 \\ 1 & 4 & 1 \end{pmatrix}\begin{pmatrix} v_x \\ v_y \\ v_z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \Leftrightarrow \begin{cases} v_x + 4v_y + v_z = 0 & (1) \\ 2v_x - v_y + 2v_z = 0 & (2) \\ v_x + 4v_y + v_z = 0 \end{cases}$$

Extracting v_z from (1) and plugging it into (2):

$$2v_x - v_y + 2(-v_x - 4v_y) = 0 \quad \Rightarrow v_y = 0$$

From (1) we find $v_x + v_z = 0$

$$\Rightarrow v_1 = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$$

$\lambda = 6$

$$\begin{pmatrix} 1 & 4 & 1 \\ 4 & -2 & 4 \\ 1 & 4 & 1 \end{pmatrix}\begin{pmatrix} v_x \\ v_y \\ v_z \end{pmatrix} = 6\begin{pmatrix} v_x \\ v_y \\ v_z \end{pmatrix} \Leftrightarrow \begin{cases} v_x + 4v_y + v_z = 6v_x & (3) \\ 2v_x - v_y + 2v_z = 3v_y & (4) \\ v_x + 4v_y + v_z = 6v_z & (5) \end{cases}$$

From (3) and (5) we find $v_x = v_z$. Plugging this into (3):

$2v_x + 4v_y = 6v_x \Rightarrow v_x = v_y = v_z$

$\Rightarrow v_2 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$

$\underline{\lambda = -4\sqrt{2}}$

$$\begin{pmatrix} 1 & 4 & 1 \\ 4 & -2 & 4 \\ 1 & 4 & 1 \end{pmatrix} \begin{pmatrix} v_x \\ v_y \\ v_z \end{pmatrix} = -6 \begin{pmatrix} v_x \\ v_y \\ v_z \end{pmatrix} \Leftrightarrow \begin{cases} v_x + 4v_y + v_z = -6v_x & (6) \\ 2v_x - v_y + 2v_z = -3v_y & (7) \\ v_x + 4v_y + v_z = -6v_z & (8) \end{cases}$$

From (6) and (8) we find $v_x = v_z$. Plugging this into (6) we find:

$v_y = \frac{-8}{4} v_x = -2v_x$

$\Rightarrow v_3 = \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix}$

(c) $B v_1 = \begin{pmatrix} 1 & -2 & -1 \\ -2 & 2 & -2 \\ -1 & -2 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} = \begin{pmatrix} 2 \\ 0 \\ -2 \end{pmatrix} = 2 \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$

$\Rightarrow v_1$ is an eigenvector of B.

$$Bv_2 = \begin{pmatrix} 1 & -2 & -1 \\ -2 & 2 & -2 \\ -1 & -2 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} -2 \\ -2 \\ -2 \end{pmatrix} = -2 \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

$\Rightarrow v_2$ is an eigenvector of B.

$$Bv_3 = \begin{pmatrix} 1 & -2 & -1 \\ -2 & 2 & -2 \\ -1 & -2 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix} = \begin{pmatrix} 4 \\ -8 \\ 4 \end{pmatrix} = 4 \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix}$$

$\Rightarrow v_3$ is an eigenvector of B.

A.26

(a) A and B are real and symmetric $\Rightarrow$ both are normal and therefore diagonalizable.

$$AB = \begin{pmatrix} 2 & 2 & -1 \\ 2 & -1 & 2 \\ -1 & 2 & 2 \end{pmatrix}\begin{pmatrix} 2 & -1 & 2 \\ -1 & 5 & -1 \\ 2 & -1 & 2 \end{pmatrix} = \begin{pmatrix} 0 & 9 & 0 \\ 9 & -9 & 9 \\ 0 & 9 & 0 \end{pmatrix}$$

$$BA = \begin{pmatrix} 2 & -1 & 2 \\ -1 & 5 & -1 \\ 2 & -1 & 2 \end{pmatrix}\begin{pmatrix} 2 & 2 & -1 \\ 2 & -1 & 2 \\ -1 & 2 & 2 \end{pmatrix} = \begin{pmatrix} 0 & 9 & 0 \\ 9 & -9 & 9 \\ 0 & 9 & 0 \end{pmatrix}$$

$\Rightarrow [A, B] = 0$

(b)
$$\begin{vmatrix} 2-\lambda & 2 & -1 \\ 2 & -1-\lambda & 2 \\ -1 & 2 & 2-\lambda \end{vmatrix} = -(2-\lambda)^2(1+\lambda) - 4 - 4 + (1+\lambda) - 4(2-\lambda) - 4(2-\lambda) = 0$$

$(-\lambda^2 + 4\lambda - 4)(1+\lambda)$

$$\Rightarrow -\lambda^3 + 3\lambda^2 + 9\lambda - 27 = 0 \Leftrightarrow (\lambda+3)(-\lambda^2+6\lambda-9) = 0$$

$$\Leftrightarrow -(\lambda+3)(\lambda-3)^2 = 0$$

The spectrum is degenerate because $\lambda = 3$ is a double root of the characteristic equation.

$\lambda = -3$

$$\begin{pmatrix} 2 & 2 & -1 \\ 2 & -1 & 2 \\ -1 & 2 & 2 \end{pmatrix}\begin{pmatrix} v_x \\ v_y \\ v_z \end{pmatrix} = -3\begin{pmatrix} v_x \\ v_y \\ v_z \end{pmatrix} \Leftrightarrow \begin{cases} 2v_x + 2v_y - v_z = -3v_x & (1) \\ 2v_x - v_y + 2v_z = -3v_y & (2) \\ -v_x + 2v_y + 2v_z = -3v_z & (3) \end{cases}$$

$(1) \Rightarrow v_z = 5v_x + 2v_y$

$$\Rightarrow \begin{cases} (2') \; 2v_x - v_y + 10v_x + 4v_y = -3v_y \\ (3') \; -v_x + 2v_y + 10v_x + 4v_y = -15v_x - 6v_y \end{cases} \Leftrightarrow \begin{cases} 12v_x = -6v_y \\ 24v_x = -12v_y \end{cases}$$

$\Rightarrow v_y = -2v_x \Rightarrow v_z = 5v_x - 4v_x = v_x$

$$\Rightarrow v_1 = \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix}$$

$\lambda = 3$

$$\begin{pmatrix} 2 & 2 & -1 \\ 2 & -1 & 2 \\ -1 & 2 & 2 \end{pmatrix} \begin{pmatrix} v_x \\ v_y \\ v_z \end{pmatrix} = 3 \begin{pmatrix} v_x \\ v_y \\ v_z \end{pmatrix} \Leftrightarrow \begin{cases} 2v_x + 2v_y - v_z = 3v_x & (4) \\ 2v_x - v_y + 2v_z = 3v_y & (5) \\ -v_x + 2v_y + 2v_z = 3v_z & (6) \end{cases}$$

$(4) \Rightarrow v_z = -v_x + 2v_y$

$$\Rightarrow \begin{matrix} (5') \\ (6') \end{matrix} \begin{cases} 2v_x - v_y - 2v_x + 4v_y = 3v_y \\ -v_x + 2v_y - 2v_x + 4v_y = -3v_x + 6v_y \end{cases} \Leftrightarrow \begin{cases} 0 = 0 \\ 0 = 0 \end{cases}$$

There are 2 linearly independent eigenvectors, obeying (4):

$$v_2 = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}, \quad v_3 = \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix}$$

(c)

$$\begin{pmatrix} 2 & -1 & 2 \\ -1 & 5 & -1 \\ 2 & -1 & 2 \end{pmatrix} v_1 = \begin{pmatrix} 2 & -1 & 2 \\ -1 & 5 & -1 \\ 2 & -1 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix} = \begin{pmatrix} 6 \\ -12 \\ 6 \end{pmatrix} = 6 \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix}$$

$$\begin{pmatrix} 2 & -1 & 2 \\ -1 & 5 & -1 \\ 2 & -1 & 2 \end{pmatrix} v_2 = \begin{pmatrix} 2 & -1 & 2 \\ -1 & 5 & -1 \\ 2 & -1 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} = 0 \cdot \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$$

$$\begin{pmatrix} 2 & -1 & 2 \\ -1 & 5 & -1 \\ 2 & -1 & 2 \end{pmatrix} v_3 = \begin{pmatrix} 2 & -1 & 2 \\ -1 & 5 & -1 \\ 2 & -1 & 2 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 3 \\ 3 \\ 3 \end{pmatrix}$$

So that v_1 and v_2 are eigenvectors, but v_3 is not. Using the approach of (A.90), we first determine the coefficients of C:

$$B v_2 = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} = 0 \cdot v_2 + 0 \cdot v_3 \Rightarrow C_{11} = C_{21} = 0$$

$$B v_3 = \begin{pmatrix} 3 \\ 3 \\ 3 \end{pmatrix} = C_{12} \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} + C_{22} \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix} \Rightarrow C_{12} = C_{22} = 3$$

$$\Rightarrow C = \begin{pmatrix} 0 & 3 \\ 0 & 3 \end{pmatrix}$$

$$0 = \begin{vmatrix} -\nu & 3 \\ 0 & 3-\nu \end{vmatrix} = \nu^2 - 3\nu \Rightarrow \nu = 0 \text{ or } \nu = 3$$

$\lambda = 0$

$$\begin{pmatrix} 0 & 3 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} d_{11} \\ d_{21} \end{pmatrix} = 0 \cdot \begin{pmatrix} d_{11} \\ d_{21} \end{pmatrix} \iff \begin{cases} 3d_{21} = 0 \\ 3d_{21} = 0 \end{cases} \implies \begin{cases} d_{11} = 1 \\ d_{21} = 0 \end{cases}$$

$\lambda = 3$

$$\begin{pmatrix} 0 & 3 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} d_{12} \\ d_{22} \end{pmatrix} = 3 \begin{pmatrix} d_{12} \\ d_{22} \end{pmatrix} \iff \begin{cases} 3d_{22} = 3d_{12} \\ 3d_{22} = 3d_{22} \end{cases} \implies d_{12} = d_{22} = 1$$

The common eigenvectors of A and B are v_1, and

$$\tilde{v}_2 = 1 \cdot v_2 + 0 \cdot v_3 = v_2$$

and

$$\tilde{v}_3 = 1 \cdot v_2 + 1 v_3 = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

A basis of common eigenvectors is

$$\mathcal{B} = \left\{ \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \right\}$$

A.27

Thm: If $\langle\gamma|\hat{T}\gamma\rangle = \langle\hat{T}\gamma|\gamma\rangle$ for any $|\gamma\rangle$, then $\langle\alpha|\hat{T}\beta\rangle = \langle\hat{T}\alpha|\beta\rangle$ for any $|\alpha\rangle$ and $|\beta\rangle$.

Proof:

Let $|\gamma\rangle = |\alpha\rangle + |\beta\rangle$, then

$\langle\gamma|\hat{T}\gamma\rangle = \langle\hat{T}\gamma|\gamma\rangle$ means

$$\langle\alpha|\hat{T}\alpha\rangle + \langle\alpha|\hat{T}\beta\rangle + \langle\beta|\hat{T}\alpha\rangle + \langle\beta|\hat{T}\beta\rangle = \langle\hat{T}\alpha|\alpha\rangle + \langle\hat{T}\alpha|\beta\rangle + \langle\hat{T}\beta|\alpha\rangle + \langle\hat{T}\beta|\beta\rangle$$

But $\langle\cdot|\hat{T}\cdot\rangle = \langle\hat{T}\cdot|\cdot\rangle$ for any input, so we are left with

$$\langle\alpha|\hat{T}\beta\rangle + \langle\beta|\hat{T}\alpha\rangle = \langle\hat{T}\alpha|\beta\rangle + \langle\hat{T}\beta|\alpha\rangle \quad (*)$$

We now do the same thing for $|\gamma\rangle = |\alpha\rangle + i|\beta\rangle$ and find

$$\langle\alpha|\hat{T}\alpha\rangle + i\langle\alpha|\hat{T}\beta\rangle - i\langle\beta|\hat{T}\alpha\rangle + \langle\beta|\hat{T}\beta\rangle = \langle\hat{T}\alpha|\alpha\rangle + i\langle\hat{T}\alpha|\beta\rangle - i\langle\hat{T}\beta|\alpha\rangle + \langle\hat{T}\beta|\beta\rangle$$

$$\Leftrightarrow \langle\alpha|\hat{T}\beta\rangle - \langle\beta|\hat{T}\alpha\rangle = \langle\hat{T}\alpha|\beta\rangle - \langle\hat{T}\beta|\alpha\rangle \quad (**)$$

So,

$(*) + (**)$: $2\langle\alpha|\hat{T}\beta\rangle = 2\langle\hat{T}\alpha|\beta\rangle$

which is what we wanted to show. QED

A.28

(a) $T^\dagger = \begin{pmatrix} 1 & 1+i \\ 1-i & 0 \end{pmatrix}^T = \begin{pmatrix} 1 & 1-i \\ 1+i & 0 \end{pmatrix} = T$

(b) $0 = \begin{vmatrix} 1-\lambda & 1-i \\ 1+i & -\lambda \end{vmatrix} = \lambda^2 - \lambda - 2 = \boxed{(\lambda+1)(\lambda-2)}$

(c) $\lambda = -1$

$$\begin{pmatrix} 1 & 1-i \\ 1+i & 0 \end{pmatrix}\begin{pmatrix} v_x \\ v_y \end{pmatrix} = -\begin{pmatrix} v_x \\ v_y \end{pmatrix} \iff \begin{cases} v_x + (1-i)v_y = -v_x \\ (1+i)v_x = -v_y \end{cases}$$

$$\Rightarrow v = \begin{pmatrix} 1 \\ -1-i \end{pmatrix}$$

To normalise it, note $|v|^2 = 1+2 = 3$, which means the normalised eigenvector is

$$\boxed{v_1 = \begin{pmatrix} 1/\sqrt{3} \\ -(1+i)/\sqrt{3} \end{pmatrix}}$$

$\lambda = 2$

$$\begin{pmatrix} 1 & 1-i \\ 1+i & 0 \end{pmatrix}\begin{pmatrix} v_x \\ v_y \end{pmatrix} = 2\begin{pmatrix} v_x \\ v_y \end{pmatrix} \iff \begin{cases} v_x + (1-i)v_y = 2v_x \\ (1+i)v_x = 2v_y \end{cases}$$

$$\Rightarrow v = \begin{pmatrix} 1 \\ \frac{1+i}{2} \end{pmatrix}$$

$$|v|^2 = 1 + \frac{2}{4} = \frac{3}{2}$$

$$\Rightarrow \boxed{v_2 = \begin{pmatrix} \sqrt{2}/\sqrt{3} \\ (1+i)/\sqrt{6} \end{pmatrix}}$$

Checking their inner product:

$$v_1 \cdot v_2 = \frac{1}{\sqrt{3}}\begin{pmatrix} 1 & -(1-i) \end{pmatrix} \cdot \frac{1}{\sqrt{6}}\begin{pmatrix} 2 \\ 1+i \end{pmatrix} = \frac{1}{3\sqrt{2}}(2-2) = 0$$

and so they are indeed orthogonal.

(d) $S^{-1} = \begin{pmatrix} \vdots & \vdots \\ v_1 & v_2 \\ \vdots & \vdots \end{pmatrix} = \begin{pmatrix} 1/\sqrt{3} & 2/\sqrt{6} \\ -\frac{1+i}{\sqrt{3}} & \frac{1+i}{\sqrt{6}} \end{pmatrix} = \frac{1}{\sqrt{6}} \begin{pmatrix} \sqrt{2} & 2 \\ -\sqrt{2}(1+i) & 1+i \end{pmatrix}$

$$\left(S^{-1}\right)^{\dagger} = \frac{1}{\sqrt{6}} \begin{pmatrix} \sqrt{2} & 2 \\ -\sqrt{2}(1-i) & 1-i \end{pmatrix}^{t}$$

$$= \frac{1}{\sqrt{6}} \begin{pmatrix} \sqrt{2} & -\sqrt{2}(1-i) \\ 2 & 1-i \end{pmatrix}$$

We check $S^{-1} \cdot \left(S^{-1}\right)^{\dagger} = \frac{1}{6} \begin{pmatrix} \sqrt{2} & 2 \\ -\sqrt{2}(1+i) & 1+i \end{pmatrix} \begin{pmatrix} \sqrt{2} & -\sqrt{2}(1-i) \\ 2 & 1-i \end{pmatrix}$

$$= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

So S indeed unitary, i.e. $S = \left(S^{-1}\right)^{\dagger}$

We want to check that it diagonalises T.

$$STS^{-1} = \frac{1}{6} \begin{pmatrix} \sqrt{2} & -\sqrt{2}(1-i) \\ 2 & 1-i \end{pmatrix} \begin{pmatrix} 1 & 1-i \\ 1+i & 0 \end{pmatrix} \begin{pmatrix} \sqrt{2} & 2 \\ -\sqrt{2}(1+i) & 1+i \end{pmatrix}$$

$$= \frac{1}{6} \begin{pmatrix} -\sqrt{2} & \sqrt{2}(1-i) \\ 4 & 2(1-i) \end{pmatrix} \begin{pmatrix} \sqrt{2} & 2 \\ -\sqrt{2}(1+i) & 1+i \end{pmatrix}$$

$$= \begin{pmatrix} -1 & 0 \\ 0 & 2 \end{pmatrix}$$

(e) $\det(T) = \det \begin{pmatrix} 1 & 1-i \\ 1+i & 0 \end{pmatrix} = -2$

$\det(STS^{-1}) = \det \begin{pmatrix} -1 & 0 \\ 0 & 2 \end{pmatrix} = -2$

$\mathrm{Tr}(T) = \mathrm{Tr} \begin{pmatrix} 1 & 1-i \\ 1+i & 0 \end{pmatrix} = 1$

$\mathrm{Tr}(STS^{-1}) = \mathrm{Tr} \begin{pmatrix} -1 & 0 \\ 0 & 2 \end{pmatrix} = 1$

A.29

(a) $\det(T) = 8 - 1 - 1 - 2 - 2 - 2 = 0$

$T_2(T) = 2 + 2 + 2 = 6$

(b)

$$0 = \begin{vmatrix} 2-\lambda & i & 1 \\ -i & 2-\lambda & i \\ 1 & -i & 2-\lambda \end{vmatrix} = (2-\lambda)^3 - 1 - 1 - (2-\lambda) - (2-\lambda) - (2-\lambda)$$

$$= -\lambda^3 + 6\lambda^2 - 9\lambda = -\lambda(\lambda^2 - 6\lambda + 9) = -\lambda(\lambda-3)^2$$

The sum of the eigenvalues is

$$0 + 3 + 3 = 6$$

which is equal $T_2(T)$. The product is

$$0 \cdot 3 \cdot 3 = 0$$

which is equal to $\det(T)$.

T in diagonal form would be $\begin{pmatrix} 0 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{pmatrix}$ (where I've assumed it can actually be diagonalised).

(c) $\lambda = 0$

$$\begin{pmatrix} 2 & i & 1 \\ -i & 2 & i \\ 1 & -i & 2 \end{pmatrix} \begin{pmatrix} v_x \\ v_y \\ v_z \end{pmatrix} = 0 \cdot \begin{pmatrix} v_x \\ v_y \\ v_z \end{pmatrix} \Leftrightarrow \begin{cases} 2v_x + iv_y + v_z = 0 & (1) \\ -iv_x + 2v_y + iv_z = 0 & (2) \\ v_x - iv_y + 2v_z = 0 & (3) \end{cases}$$

$(1) + (3)$: $3v_x + 3v_z = 0 \Leftrightarrow v_z = -v_x$

Using this in (1) gives

$v_x + iv_y = 0 \Leftrightarrow v_y = iv_x$

$\Rightarrow v = \begin{pmatrix} 1 \\ i \\ -1 \end{pmatrix}$ and therefore $|v| = \sqrt{3}$

A normalised eigenvector is $v_1 = \boxed{\frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ i \\ -1 \end{pmatrix}}$

$\lambda = 3$

$$\begin{pmatrix} 2 & i & 1 \\ -i & 2 & i \\ 1 & -i & 2 \end{pmatrix} \begin{pmatrix} v_x \\ v_y \\ v_z \end{pmatrix} = 3 \begin{pmatrix} v_x \\ v_y \\ v_z \end{pmatrix} \Leftrightarrow \begin{cases} 2v_x + iv_y + v_z = 3v_x \\ -iv_x + 2v_y + iv_z = 3v_y \\ v_x - iv_y + 2v_z = 3v_z \end{cases}$$

$$\begin{cases} -v_x + iv_y + v_z = 0 \\ -iv_x - v_y + iv_z = 0 \\ v_x - iv_y - v_z = 0 \end{cases} \Rightarrow v_z = v_x - iv_y$$

All three equations are equivalent. So two linearly independent solutions are

$$w_1 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \text{ and } w_2 = \begin{pmatrix} 0 \\ 1 \\ -i \end{pmatrix}$$

To orthogonalise the solutions we look for α and β such that $\alpha w_1 + \beta w_2$ is orthogonal to w_1:

$$\begin{pmatrix} \alpha \\ \beta \\ \alpha - i\beta \end{pmatrix}$$

$$(1 \ 0 \ 1) \begin{pmatrix} \alpha \\ \beta \\ \alpha - i\beta \end{pmatrix} = \alpha + \alpha - i\beta = 2\alpha - i\beta = 0 \Rightarrow \beta = -2i\alpha$$

So a solution is $\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} - 2i \begin{pmatrix} 0 \\ 1 \\ -i \end{pmatrix} = \begin{pmatrix} 1 \\ -2i \\ -1 \end{pmatrix}$. We found orthogonal eigenvectors with $\lambda = 3$:

$$\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \text{ and } \begin{pmatrix} 1 \\ -2i \\ -1 \end{pmatrix}$$

Normalising them we get

$$v_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \text{ and } v_3 = \frac{1}{\sqrt{6}} \begin{pmatrix} 1 \\ -2i \\ -1 \end{pmatrix}$$

$$\langle v_1 | v_2 \rangle = \frac{1}{\sqrt{6}} (1 \ -1) = 0$$

$$v_1 = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ i \\ -1 \end{pmatrix}$$

$$\langle v_1 | v_3 \rangle = \frac{1}{3\sqrt{2}} [1 - 2 + 1] = 0$$

(d)

$$S^{-1} = \frac{1}{\sqrt{6}} \begin{pmatrix} \sqrt{2} & \sqrt{3} & 1 \\ i\sqrt{2} & 0 & -2i \\ -\sqrt{2} & \sqrt{3} & -1 \end{pmatrix}$$

Because the columns of S^{-1} are orthonormal, it is unitary. So

$$S = (S^{-1})^\dagger = \frac{1}{\sqrt{6}} \begin{pmatrix} \sqrt{2} & \sqrt{3} & 1 \\ -i\sqrt{2} & 0 & 2i \\ -\sqrt{2} & \sqrt{3} & -1 \end{pmatrix}^t = \frac{1}{\sqrt{6}} \begin{pmatrix} \sqrt{2} & -i\sqrt{2} & -\sqrt{2} \\ \sqrt{3} & 0 & \sqrt{3} \\ 1 & 2i & -1 \end{pmatrix}$$

$$STS^{-1} = \frac{1}{6}\begin{pmatrix} \sqrt{2} & -i\sqrt{2} & -\sqrt{2} \\ \sqrt{3} & 0 & \sqrt{3} \\ 1 & 2i & -1 \end{pmatrix}\begin{pmatrix} 2 & i & 1 \\ -i & 2 & i \\ 1 & -i & 2 \end{pmatrix}\begin{pmatrix} \sqrt{2} & \sqrt{3} & 1 \\ i\sqrt{2} & 0 & -2i \\ -\sqrt{2} & \sqrt{3} & -1 \end{pmatrix}$$

$$= \frac{1}{6}\begin{pmatrix} 0 & 0 & 0 \\ 3\sqrt{3} & 0 & 3\sqrt{3} \\ 3 & 6i & -3 \end{pmatrix}\begin{pmatrix} \sqrt{2} & \sqrt{3} & 1 \\ i\sqrt{2} & 0 & -2i \\ -\sqrt{2} & \sqrt{3} & -1 \end{pmatrix}$$

$$= \begin{pmatrix} 0 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{pmatrix}$$

which is the diagonal form of T.

A.30

(a) $\langle \hat{u}\alpha | \hat{u}\beta \rangle = \langle \hat{u}^\dagger \hat{u}\alpha | \beta \rangle = \langle \alpha | \beta \rangle$

(b) Let $\hat{u}|\alpha\rangle = \lambda|\alpha\rangle$ with $|\alpha\rangle \neq 0$

$\Rightarrow \langle \hat{u}\alpha | \hat{u}\alpha \rangle = |\lambda|^2 \langle \alpha | \alpha \rangle$

But applying the result of (a), we also have

$\langle \hat{u}\alpha | \hat{u}\alpha \rangle = \langle \alpha | \alpha \rangle$

Therefore, $|\lambda|^2 = 1$

(c) Let $\hat{u}|\alpha\rangle = \lambda|\alpha\rangle$ and $\hat{u}|\beta\rangle = \mu|\beta\rangle$, with $\lambda \neq \mu$ and $|\alpha\rangle$ and $|\beta\rangle$ both non-zero. From (b) we also know that λ and μ are non-zero. Because $\hat{u}|\beta\rangle = \mu|\beta\rangle$,

$\hat{u}^\dagger(\hat{u}|\beta\rangle) = \hat{u}^\dagger(\mu|\beta\rangle)$

$\Leftrightarrow |\beta\rangle = \mu \hat{u}^\dagger|\beta\rangle \Leftrightarrow \hat{u}^\dagger|\beta\rangle = \frac{1}{\mu}|\beta\rangle$

Using this we write $\langle \beta | \hat{u}\alpha \rangle$ in two different ways

$\langle \beta | \hat{u}\alpha \rangle = \lambda \langle \beta | \alpha \rangle$ (*)

but also

$\langle \beta | \hat{u}\alpha \rangle = \langle u^\dagger \beta | \alpha \rangle = \frac{1}{\mu^*} \langle \beta | \alpha \rangle = \mu \langle \beta | \alpha \rangle$ (**)

where the last step follows from (b) via

$1 = |\mu|^2 = \mu\mu^*$

Taking (*) and (**) together, with $\lambda \neq \mu$, gives

$\langle \beta | \alpha \rangle = 0$

A.31

$$e^M := \sum_{k=0}^{\infty} \frac{M^k}{k!}$$

(a) (i) $M = \begin{pmatrix} 0 & 1 & 3 \\ 0 & 0 & 4 \\ 0 & 0 & 0 \end{pmatrix} \Rightarrow M^2 = \begin{pmatrix} 0 & 1 & 3 \\ 0 & 0 & 4 \\ 0 & 0 & 0 \end{pmatrix}\begin{pmatrix} 0 & 1 & 3 \\ 0 & 0 & 4 \\ 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 4 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$

$$M^3 = \begin{pmatrix} 0 & 0 & 4 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}\begin{pmatrix} 0 & 1 & 3 \\ 0 & 0 & 4 \\ 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

and therefore, $M^k = 0$ for all $k \geq 3$.

$$\Rightarrow e^M = I + M + \frac{1}{2}M^2 = \begin{pmatrix} 1 & 1 & 5 \\ 0 & 1 & 4 \\ 0 & 0 & 1 \end{pmatrix}$$

(ii) $M = \begin{pmatrix} 0 & \theta \\ -\theta & 0 \end{pmatrix} = \theta\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$

$$M^2 = \theta^2\begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} = -\theta^2 I$$

$$M^3 = -\theta^3\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

$$M^4 = +\theta^4 I$$

$$M^5 = \theta^5\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

$$\Rightarrow M^k = \begin{cases} (-1)^{\frac{k-1}{2}}\theta^k \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} & (k \text{ odd}) \\ (-1)^{\frac{k}{2}}\theta^k I & (k \text{ even}) \end{cases}$$

So we have

$$\sum_{k=0}^{\infty} \frac{M^k}{k!} = I + \theta\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} - \frac{1}{2!}\theta^2 I - \frac{1}{3!}\theta^3\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} + \frac{1}{4!}\theta^4 I + \frac{1}{5!}\theta^5\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} + \cdots$$

$$= I\left(1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \cdots\right) + \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}\left(\theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \cdots\right)$$

$$= \cos\theta\, I + \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}\sin\theta$$

$$= \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix}$$

(b) Thm: If M is diagonalizable, then $\det(e^M) = e^{\mathrm{Tr}(M)}$

Proof: Let $\lambda_1, \ldots, \lambda_n$ be the eigenvalues (not necessarily distinct) of M. And let S be the transformation matrix such that

$$D = \begin{pmatrix} \lambda_1 & & 0 \\ & \lambda_2 & \\ 0 & & \ddots \\ & & & \lambda_n \end{pmatrix} = SMS^{-1}$$

Then

$$M^k = \left(S^{-1} D S\right)^k = S^{-1} D^k S$$

$$\Rightarrow e^M = \sum_{k=0}^{\infty} \frac{M^k}{k!} = S^{-1} \left(\sum_{k=0}^{\infty} \frac{D^k}{k!} \right) S$$

In the resulting matrix in the parentheses, the off-diagonal entries are 0, and the diagonal ones are (for the j-th row)

$$1 + \lambda_j + \frac{1}{2!}\lambda_j^2 + \frac{1}{3!}\lambda_j^3 + \cdots = e^{\lambda_j}$$

$$\Rightarrow e^M = S^{-1} \begin{pmatrix} e^{\lambda_1} & & 0 \\ & e^{\lambda_2} & \\ 0 & & \ddots \\ & & & e^{\lambda_n} \end{pmatrix} S$$

$$\Rightarrow \det(e^M) = \det(S^{-1}) \cdot \det \begin{pmatrix} e^{\lambda_1} & & 0 \\ & \ddots & \\ 0 & & e^{\lambda_n} \end{pmatrix} \det(S) = e^{\lambda_1 + \cdots + \lambda_n}$$

$$= e^{\mathrm{Tr}(M)}$$

QED

(c)

Thm: If $[M, N] = 0$, then $e^{M+N} = e^M e^N$

proof:

Let $MN = NM$

$$e^{M+N} = \sum_{k=0}^{\infty} \frac{(M+N)^k}{k!}$$

$$= \sum_{k=0}^{\infty} \frac{1}{k!} \left(\sum_{\ell=0}^{k} \binom{k}{\ell} M^{\ell} N^{k-\ell} \right)$$

where the last equality holds because $MN = NM$.

$$e^{M+N} = \sum_{k=0}^{\infty} \sum_{\ell=0}^{k} \frac{1}{k!} \binom{k}{\ell} M^{\ell} N^{k-\ell} = \sum_{\ell=0}^{\infty} \sum_{k=\ell}^{\infty} \frac{1}{\ell!\,(k-\ell)!} M^{\ell} N^{k-\ell}$$

Renaming $k - \ell$ as t, this becomes

$$e^{M+N} = \sum_{\ell=0}^{\infty} \sum_{t=0}^{\infty} \frac{1}{\ell!\, t!} M^{\ell} N^{t} = \left(\sum_{\ell=0}^{\infty} \frac{M^{\ell}}{\ell!} \right) \left(\sum_{t=0}^{\infty} \frac{N^{t}}{t!} \right) = e^M e^N$$

QED

To show that $[M,N]=0$ is required, we choose

$M=\begin{pmatrix}1&0\\0&0\end{pmatrix}$ and $N=\begin{pmatrix}0&0\\1&0\end{pmatrix}$

Then

$MN=\begin{pmatrix}0&0\\0&0\end{pmatrix}$ but $NM=\begin{pmatrix}0&0\\1&0\end{pmatrix}$

$M^k = M$ for all $k \geq 1$, and

$N^k = 0$ for all $k > 1$

Therefore $e^M = \sum_{k=0}^{\infty} \frac{M^k}{k!} = I + M\sum_{k=1}^{\infty} \frac{1}{k!} = I + (e-1)M$

$= \begin{pmatrix}e&0\\0&1\end{pmatrix}$

$$e^N = \sum_{k=0}^{\infty} \frac{N^k}{k!} = I + N = \begin{pmatrix}1&0\\1&1\end{pmatrix}$$

$\Rightarrow e^M e^N = \begin{pmatrix}e&0\\1&1\end{pmatrix}$

$M+N = \begin{pmatrix}1&0\\1&0\end{pmatrix}$, so that

$(M+N)^k = M+N$ for all $k \geq 1$

Therefore

$$e^{M+N} = \sum_{k=0}^{\infty} \frac{(M+N)^k}{k!} = I + (M+N)(e-1) = \begin{pmatrix}e&0\\e-1&1\end{pmatrix} \neq e^M e^N$$

(d)

Thm: Let $H^\dagger = H$. Then $\left(e^{iH}\right)^\dagger = \left(e^{iH}\right)^{-1}$

Proof:

$$\left(e^{iH}\right)^\dagger = \left(\sum_{k=0}^{\infty} \frac{(iH)^k}{k!}\right)^\dagger = \sum_{k=0}^{\infty} \frac{\left[(iH)^\dagger\right]^k}{k!} = \sum_{k=0}^{\infty} \frac{(-iH)^k}{k!} = e^{-iH}$$

Because $[iH, -iH]$ commute, and using the theorem in (c), we find

$$e^{iH}\cdot\left(e^{iH}\right)^\dagger = e^{iH}\cdot e^{-iH} = e^0 = I$$

and thus

$$\left(e^{iH}\right)^{\dagger} = \left(e^{iH}\right)^{-1}$$

QED